

THE ONE SQUARE KILOMETRE PROBLEM



Cremorne packs more than 700 businesses and over 10,000 workers into less than a square kilometre, on streets and public space built for manufacturing.¹ Public open space runs to roughly two square metres per person, the lowest provision in Melbourne, and that count excludes the ten thousand people who arrive each weekday.² Footpaths are narrow, commuter traffic and service vehicles compete for the same streets, and there is little canopy against an afternoon in February.³

The precinct is also data-dark. Nobody can say how busy Cremorne is right now, where the crush is, which spaces are sitting empty, or how hot the walk up from the station has become. Every building is run on its own, so 700 tenants solve the same problems separately and none of them can see the whole.

Build the software, data or sensing layer that lets Cremorne be operated as one system rather than 700 separate tenancies, and use it to measurably improve the working day.

We are after working technology, not a planning submission. Assume no new land, no new station and no new buildings. Show us a running prototype, what it measures, and who it does not serve.

■ SPONSOR CONTEXT

Cremorne Digital Hub sits inside Victoria's densest technology precinct. Cremorne is smaller than one square kilometre and holds more than 700 businesses and over 10,000 workers, including REA Group, SEEK, MYOB and carsales. It contributed close to \$4 billion to the Victorian economy in 2019.¹ Our job is to connect that concentration of industry with the wider innovation ecosystem, which means we care a great deal about whether the precinct is somewhere people actually want to be.

■ THE PROBLEM

Cremorne was built for manufacturing and is now running an innovation economy on top of streets, drainage and public space that were never designed for it. The City of Yarra's own built form review puts public open space at roughly two square metres per person, the lowest provision in Melbourne, and that figure counts residents rather than the ten thousand people who arrive each weekday.² Footpaths are narrow, the streets carry commuter traffic and service vehicles at the same time, and there is very little tree canopy to take the heat out of a summer afternoon.³

For most of the precinct's history this did not matter much, because coming in was not optional. It is now. Hybrid work has turned physical presence into a choice that people make, or decline, one day at a time. A precinct that is unpleasant to walk around, difficult to reach and empty after six is competing against a spare room at home, and it is not obviously winning.

What makes this tractable rather than merely expensive is that Cremorne is data-dark. For a precinct full of technology companies, remarkably little is known about how it actually runs. There is no live picture of crowding, heat, noise or spare capacity. Occupancy is measured building by building, if at all. Nobody can answer basic questions like where the pinch points are at 8:45am, which meeting rooms sit empty four days in five, or how much hotter the Church Street walk is than the river path. The physical constraints are fixed for a decade. The information layer is not.

■ THE CHALLENGE

Build the software, data or sensing layer that lets Cremorne be operated as one system rather than 700 separate tenancies, and use it to measurably improve the working day.

Your solution can be software, hardware, data, or a combination of the three. We are a digital hub and we want working technology. Planning submissions, policy papers and concept decks are not what we are after.

Assume no new land, no new station and no new buildings. Those are being argued about on a ten to twenty year timescale.⁴ We are interested in what can be built inside the existing square kilometre, with existing assets, in the next twelve months.

SOURCES

- ¹ Victorian Government, *Plan to Cement Cremorne as a Tech, Jobs Powerhouse*, December 2020. premier.vic.gov.au/plan-cement-cremorne-tech-jobs-powerhouse
- ² Hodyl & Co for City of Yarra, *Cremorne Built Form Review*, May 2022. Open space provision of approximately 2 sq m per person, the lowest in Melbourne, excluding roughly 10,000 additional weekday arrivals. yarracity.vic.gov.au
- ³ City of Yarra, *Cremorne Streets and Movement Strategy*, June 2020. yarracity.vic.gov.au
- ⁴ Victorian Planning Authority with City of Yarra, *Cremorne Place Implementation Plan*, December 2020. vpa.vic.gov.au/project/cremorne